University of Cambridge Mathematical Tripos Part III

Analysis of PDEs

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Course description

This course serves as an introduction to the mathematical study of Partial Differential Equations (PDEs). The theory of PDEs is presently a major area of active research, and it goes back to the very birth of mathematical analysis in the 18th and 19th centuries. The subject lies at the crossroads of physics and many areas of pure and applied mathematics. The course will mostly focus on modern techniques, with emphasis on functional analytic techniques, relying on a priori estimates rather than explicit solutions. The course will primarily focus on approaches to linear, elliptic and evolutionary problems through weak formulations and energy estimates, with the prototypical examples being Laplace's equation, the heat equation, the transport equation, the wave equation and Schrödinger's equation. The following topics will be studied.

- *Introduction and Cauchy–Kovalevskaya theorem.* Definition and examples of PDEs; wellposed problems; classification; real analyticity; Cauchy–Kovalevskaya theorem.
- *Spaces of functions.* Hölder spaces; weak derivatives; Sobolev spaces; approximation by mollifiers; extension and trace theorems; Sobolev embeddings.
- *Elliptic boundary value problems.* Weak formulation; solvability via Lax–Milgram; Fredholm alternative for elliptic PDEs and its consequences on the spectrum; regularity of solutions via the difference quotient method.
- *Hyperbolic equations.* Notion of hyperbolicity for second-order linear operators; weak formulation of the initial boundary value problem; existence via Galerkin's method; uniqueness via the vector field method; finite speed of propagation; first-order hyperbolic PDEs and the method of characteristics.

Prerequisites

There are no specific pre-requisites beyond a standard undergraduate analysis background, in particular a familiarity with measure theory and integration. The course will be mostly self-contained and can be used as a first introductory course in PDEs for students wishing to continue with some specialised PDE Part III courses in the Lent and Easter terms.

Contents

1 Introduction **3**

1.1 Overview 3

1.2 Classification of PDEs 4

1.3 Some lessons from ODEs 4

1.4 Data and well-posedness 5

2 Cauchy–Kovalevskaya theorem **5**

2.1 Real analyticity and majorants 5

2.2 Cauchy–Kovalevskaya for ODEs 7

2.3 Cauchy–Kovalevskaya for PDEs 8

1 Introduction

1.1 Overview

An ordinary differential equation (ODE) is an equation of the form

$$F(x, u(x), u'(x), \cdots, u^{(k)}(x)) = 0, \qquad x \in I$$

where $u \in C^k(I)$ and $I \subset \mathbb{R}$ is an interval. Partial differential equations (PDEs) arise when u depends on several real variables, $u = u(x_1, \cdots, x_n)$ for $n \geq 2$. We work on a *domain*, i.e. an open, non-empty, connected subset $\mathcal{U} \subset \mathbb{R}^n$.

Definition Partial differential equation
Let $n \geq 2$. A PDE of order k is an equation

$$F(x, u(x), Du(x), \cdots, D^k u(x)) = 0, \qquad x \in \mathcal{U} \tag{1}$$

where $u \colon \mathcal{U} \to \mathbb{R}$ is the unknown and $F \colon \mathcal{U} \times \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^{n^2} \times \cdots \times \mathbb{R}^{n^k} \to \mathbb{R}$ a given function.

Remarks

- We can also have $u \in \mathbb{R}^p$ and $F \in \mathbb{R}^q$, yielding a system of PDEs.
- If $x_1 = t$, then we sometimes call the PDE *evolutionary*.

Definition Classical solution
A classical solution to the PDE (1) *is a function $u \in C^k(\mathcal{U})$ for which* (1) *holds pointwise on $\mathcal{U}$.*

Notation. $u_x = \partial_x u = \frac{\partial u}{\partial x}$, $Du = (\partial_{x_i} u)_i = \nabla u$, $D^2 u = (\partial^2_{x_i x_j} u)_{ij}$, etc.

Examples
Many partial differential equations arise from mathematical modelling in applied mathematics.

- SIR model

$$\dot{u}(t) = f(u, v)$$
$$\dot{v}(t) = g(u, v)$$

- Heat/diffusion equation

$$u_t = d\nabla u$$

  where d is the diffusion coefficient and $\Delta = \sum_i \partial^2_{x_i}$ is the Laplace operator.
- Reactor-diffusion equation

$$u_t = \nabla u + f(u, v)$$
$$v_t = \nabla v + g(u, v)$$

- Transport equation

$$u_t + \langle v, Du \rangle = 0$$

- Poisson equation

$$\nabla u = f(x)$$

  If $f = 0$, then this could describe an electrostatic potential. If $f \neq 0$, then this could describe a Newtonian gravitational potential.
- Euler equation and Navier–Stokes equations
- Maxwell's equations and the wave equation (hyperbolic PDEs)

$$\nabla \cdot \mathbf{E} = \rho, \qquad \nabla \cdot \mathbf{B} = 0, \qquad \partial_t = \nabla \times \mathbf{B} - \mathbf{J}, \qquad \partial_t \mathbf{B} = -\nabla \times \mathbf{E}$$

- Schrödinger equation, KdV equation (dispersive PDEs)

$$iu_t = -\nabla u + Vu$$

PDEs can also arise in pure mathematics:

- Cauchy–Riemann equations
- Ricci flow

$$\partial_t g = -2\,\mathrm{Ric}_g$$

All of the equations in the examples above need additional boundary information to solve, e.g. $u|_{\partial\mathcal{U}}$ for the Laplace equation. Such boundary information is the *data* of the problem. There is no single theory for PDEs — one instead exploits structure (elliptic/hyperbolic/parabolic) and other properties (scaling symmetry, invariances, types of data, etc.). In this course, we take a modern approach to PDEs in the sense that we study existence, uniqueness and other qualitative properties instead of finding explicit representation formulae for the solutions.

1.2 Classification of PDEs

Notation. In this course, we will use *multi-index notation*. For a *multi-index* $\alpha = (\alpha_1, \cdots, \alpha_n) \in \mathbb{N}^n$ of *order* $|\alpha| = \alpha_1 + \cdots + \alpha_n$, we define

$$D^\alpha f(x) := \frac{\partial^{|\alpha|}}{\partial_{x^1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n}} f(x)$$

and

$$x^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}, \qquad \alpha! := \alpha_1! \cdots \alpha_n!$$

Definition Linear

We say that (1) *is linear if F is linear in u and its derivatives, i.e.*

$$\sum_{\alpha\colon |\alpha| \le k} a_\alpha(x) D^\alpha u(x) = f(x)$$

If $f \equiv 0$, we say that the PDE is homogeneous.

Definition Semilinear

We say that (1) *is semilinear if the highest-order derivatives appear linearly with coefficients depending only on x, i.e.*

$$\sum_{\alpha\colon |\alpha| = k} a_\alpha(x) D^\alpha u(x) + a_0(x, u, \cdots, D^{k-1}u) = 0$$

Definition Quasilinear

We say that (1) *is quasilinear if the highest-order derivatives appear linearly with coefficients depending on x and the lower-order derivatives, i.e.*

$$\sum_{\alpha\colon |\alpha| = k} a_\alpha(x, u, Du, \cdots, D^{k-1}u) D^\alpha u(x) + a_0(x, u, \cdots, D^{k-1}u) = 0$$

Definition Fully nonlinear

We say that (1) *is fully nonlinear if it is not quasilinear.*

1.3 Some lessons from ODEs

Consider the ODE

$$\begin{cases} u'(t) = f(t, u(t)) \\ u(t_0) = f_0 \end{cases} \tag{2}$$

The goal is to show existence and uniqueness for solutions depending on the regularity of f. There are roughly three key theorems that establish this.

Theorem 1.1 Cauchy–Kovalevskaya for ODEs
Suppose f is a real analytic function on a neighbourhood of (t_0, u_0). Then there exists a unique real analytic solution of (2) defined on a neighbourhood of t_0.

Theorem 1.2 Cauchy–Lipschitz / Picard–Lindelöf
Suppose f is continuous in both variables and Lipschitz in the second variable u. Then there exists a unique C^1 solution to (2) defined on a neighbourhood of t_0.

Remark. By considering the ODE $u'(t) = u(t)^2, u_0 > 0$, the solution to this ODE given by the above theorem is only guaranteed to hold *locally*.

Theorem 1.3 Cauchy–Peano
Suppose f is continuous. Then (2) admits at least one C^1 solution.

Remark. By considering the ODE $u'(t) = \sqrt{u(t)}, u_0 = 0$, we indeed see that solutions given by the above theorem need not be unique.

1.4 Data and well-posedness

Definition PDE problem
A PDE problem consists of a PDE, a domain $\mathcal{U}$, and some boundary data.

Definition Well-posedness
Let X be a function space. We say that a PDE problem is well-posed in X if
(i) *there exists a solution $u \in X$ to the PDE that agrees with the boundary data*
(ii) *the solution u is unique in X*
(iii) *the solution u depends continuously on the initial data (in a suitable topology)*

Remarks
- (i) and (ii) imply that X needs to be large enough to admit a solution but small enough for the solution to be unique
- (iii) is often motivated by physical considerations

2 Cauchy–Kovalevskaya theorem

2.1 Real analyticity and majorants

Definition Real analytic
Let $\mathcal{U} \subset \mathbb{R}^n$ be open. We say that a function $f\colon \mathcal{U} \to \mathbb{R}$ is real analytic near $\tilde{x} \in \mathcal{U}$ if there exists $r > 0$ and constants $f_\alpha \in \mathbb{R}$ such that

$$f(x) = \sum_{\alpha \in \mathbb{N}^n} f_\alpha (x - \tilde{x})^\alpha, \qquad x \in B_r(\tilde{x})$$

We then say that $f\colon \mathcal{U} \to \mathbb{R}$ is real analytic on $\mathcal{U}$ if it is real analytic near every $\tilde{x} \in \mathcal{U}$.

Notation. We denote the set of real analytic functions on $\mathcal{U}$ by $C^\omega(\mathcal{U})$.

Remark. If $f \in C^\omega(\mathcal{U})$, then $f \in C^\infty(\mathcal{U})$ and in fact

$$f_\alpha = \frac{D^\alpha f(\tilde{x})}{\alpha!}$$

Examples

- Polynomials, exp, sin, cos are real analytic on $\mathbb{R}$
- x^{-1} and $\sqrt{x}$ are real analytic on $\{x > 0\}$
- The map
$$f(x) = \begin{cases} e^{-1/x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$
is smooth but not real analytic. The same holds for every $f \in \mathcal{C}_c^\infty$.
- Given $r > 0$, consider the geometric series
$$f(x) = \frac{r}{r - (x_1 + \cdots + x_n)}, \qquad |x| < \frac{r}{\sqrt{n}}$$
Note that we can rewrite this as
$$f(x) = \sum_{k \geq 0} \left(\frac{x_1 + \cdots + x_n}{r} \right)^k$$
since
$$|x_1 + \cdots + x_n| \leq |x|\sqrt{n} < r$$
By the multinomial theorem (cf. Example Sheet 1), we have
$$f(x) = \sum_{k \geq 0} \frac{1}{r^k} \left(\sum_{|\alpha| = k} \binom{|\alpha|}{\alpha} x^\alpha \right), \qquad \binom{|\alpha|}{\alpha} = \frac{|\alpha|!}{\alpha!}$$
Thus, we have
$$f(x) = \sum_\alpha f_\alpha x^\alpha, \qquad f_\alpha = \frac{|\alpha|!}{\alpha!} \frac{1}{r^{|\alpha|}}$$
It is easy to check that the series converges on a neighbourhood of 0.

Proposition 2.1

Let $\mathcal{U} \subset \mathbb{R}^n$ be open. A function $f \colon \mathcal{U} \to \mathbb{R}$ is real analytic iff

(i) $f \in C^\infty(\mathcal{U})$

(ii) *for every compact $K \subset \mathcal{U}$, there exist constants $C(K), r(K) > 0$ such that*
$$|D^\alpha f(x)| \leq C(K) \frac{\alpha!}{r(K)^{|\alpha|}}$$
for every α and every $x \in K$

Proof. See Example Sheet 1. ∎

Definition Majorant

Let $f = \sum_\alpha f_\alpha x^\alpha$ and $g = \sum_\alpha g_\alpha x^\alpha$ be two power series. We say that g majorises f, or that g is a majorant of f, if $g_\alpha \geq |f_\alpha|$ for every α. We then write $g \gg f$.

Lemma 2.2 Comparison properties of majorants

(i) *If $g \gg f$ and g converges for $|x| < r$, then f converges absolutely for $|x| < r$*

(ii) *If f converges absolutely for $|x| < r$ and $s \in (0, r/n)$, then there exists a majorant of f that converges for $|x| < s$.*

Proof. (i): Fix $x \in B_r(0)$ and set $y = (|x_1|, \cdots, |x_n|)$. Then $|y| = |x| < r$ and the truncated series is bounded above by
$$\sum_{|\alpha| \leq k} |f_\alpha x^\alpha| = \sum_{|\alpha| \leq k} |f_\alpha| y^\alpha \leq \sum_{|\alpha| \leq k} g_\alpha y^\alpha \leq \sum_\alpha g_\alpha y^\alpha = g(y)$$

Hence, f converges absolutely at x.

(ii): Fix $s \in (0, r/n)$ and pick $\hat{s} \in (\sqrt{n}s, r/\sqrt{n})$. Set $y = (\tilde{s}, \cdots, \tilde{s})$. Then $|y| = \hat{s}\sqrt{n} < r$. By absolute convergence of f, there exists a constant C such that $|f_\alpha y^\alpha| \leq C$ for every α. Then

$$|f_\alpha| \leq \frac{C}{|y^\alpha|} = \frac{C}{|y_1|^{\alpha_1} \cdots |y_n|^{\alpha_n}} = \frac{C}{\hat{s}^{\alpha_1 + \cdots + \alpha_n}} = \frac{C}{\hat{s}^{|\alpha|}} \leq \frac{C}{\hat{s}^{|\alpha|}} \frac{|\alpha|!}{\alpha!}$$

We then define

$$g(x) = \frac{C\hat{s}}{\hat{s} - (x_1 + \cdots + x_n)} = C \sum_\alpha \frac{|\alpha|!}{\hat{s}^{|\alpha|}\alpha!} x^\alpha$$

By the geometric series example given above, this converges for $|x| < s < \hat{s}/\sqrt{n}$. ■

Remark. If $\mathcal{U} \subset \mathbb{R}^n$ is a non-empty, connected, open subset, then $f \in C^\omega(\mathcal{U})$ is uniquely determined by $\{D^\alpha f(\tilde{x}) \colon \alpha \in \mathbb{N}_n\}$ for a given $\tilde{x} \in \mathcal{U}$. This will be proven in Example Sheet 1.

2.2 Cauchy–Kovalevskaya for ODEs

Theorem 2.3 Cauchy–Kovalevskaya for ODEs
Fix $a, b > 0$ and $u_0 \in \mathbb{R}$. Let $f \colon (u_0 - b, u_0 + b) \to \mathbb{R}$ be real analytic. If $u \colon (-a, a) \to (u_0 - b, u_0 + b)$ is a C^1 solution of

$$\begin{cases} u'(t) = f(u(t)) \\ u(0) = u_0 \end{cases}$$

then u is real analytic on $(-a, a)$.

Proof. WLOG, set $u_0 = 0$. Given that u is a C^1 solution to $u'(t) = f(u(t))$, then the fact that f is C^1 implies that $u' \in C^1$ and thus $u \in C^2$. Proceeding inductively, we deduce that $u \in C^\infty$.

Next, we construct the Taylor series coefficients:

$$\begin{aligned} u^{(1)}(t) &= f^{(0)}(u(t)) \\ u^{(2)}(t) &= f^{(1)}(u(t))u^{(1)}(t) = f^{(1)}(u(t))f^{(0)}(u(t)) \\ u^{(3)}(t) &= f^{(2)}(u(t))(f^{(0)}(u(t)))^2 + (f^{(1)}(u(t)))^2 f^{(0)}(u(t)) \\ &\vdots \end{aligned}$$

Inductively, we find that

$$u^{(n)}(0) = \mathcal{P}_n(f^{(0)}(u(0)), f^{(1)}(u(0)), \cdots, f^{(n-1)}(u(0)))$$

for some polynomial $\mathcal{P}_n$ with non-negative coefficients that do not depend on the choice of f. We then define

$$\hat{u}(t) = \sum_{n \in \mathbb{Z}_{\geq 0}} \frac{1}{n!} \mathcal{P}_n(f^{(0)}(u(0)), \cdots, f^{(n-1)}(u(0)))t^n$$

with $\mathcal{P}_0 = u(0) = u_0 = 0$.

To show that u is real analytic, we proceed through the *method of majorants*. By Proposition 2.1, there exist $C, r > 0$ such that

$$|f^{(n)}(0)| \leq C\frac{n!}{r^n}$$

for every $n \in \mathbb{Z} \geq 0$. We then consider the real analytic function

$$g(x) = \sum_{n \in \mathbb{Z}_{\geq 0}} C\frac{x^n}{r^n} = \frac{Cr}{r - x}$$

for $|x| < r$. By construction, we have $g^{(n)}(0) = Cn!r^{-n} \geq |f^{(n)}(0)|$, so g majorises f at 0. Notably, the auxiliary IVP

$$\begin{cases} v'(t) = g(v(t)) \\ v(0) = 0 \end{cases}$$

admits the explicit solution

$$v(t) = r - r\sqrt{1 - \frac{2Ct}{r}}$$

which is real analytic for $|t| < R = \frac{r}{2C}$. By the above scheme (for u), we can similarly obtain the Taylor series coefficients of v as

$$v^{(n)}(0) = \mathcal{P}_n(g^{(0)}(0), \cdots, g^{(n-1)}(0))$$

Since g majorises f at 0, we have that

$$\begin{aligned} |u^{(n)}(0)| &\leq \mathcal{P}_n(|f^{(0)}(u(0))|, \cdots, |f^{(n-1)}(u(0))|) \\ &\leq \mathcal{P}_n(g^{(0)}(0), \cdots, g^{(n-1)}(0)) \\ &= v^{(n)}(0) \end{aligned}$$

Since v is real analytic on $\{|t| < R\}$ and $v \gg \hat{u}$, it follows from Lemma 2.2 that $\hat{u}$ is real analytic near 0. Then $\hat{u}'$ and $f \circ \hat{u}$ are also real analytic and all their derivatives agree at 0. Hence, $\hat{u}$ solves the ODE on a neighbourhood of 0. By uniqueness, $u = \hat{u}$ near 0 and is, in particular, real analytic near 0. ∎

Remarks

- The above proof can be extended to a system of ODEs by considering the vector majorant $(g_j)_{j=1}^n$ with
$$g_1 = \cdots = g_n = \frac{Cr}{r - (x_1 + \cdots + x_n)}$$
- To deal with non-autonomous ODEs (i.e. $f = f(t, u(t))$), we can reduce it to an autonomous ODE for $w(t) = (t, u(t))$.

2.3 Cauchy–Kovalevskaya for PDEs

Consider the first-order quasilinear PDE system

$$\begin{cases} u_{x_n} = \sum_{j=1}^{n-1} B_j(u, x')u_{x_j} + C(u, x') \\ u(x', x_n = 0) = 0, \qquad x' \in B_r(0) \subset \mathbb{R}^{n-1} \end{cases}$$

for some unknown $u \colon B_r^n \to \mathbb{R}^m$.

Theorem 2.4 Cauchy–Kovalevskaya for first-order PDEs
Let $\varrho > 0$. If $(B_j)_{j=1}^{n-1}, C$ are real analytic on $\{|u|^2 + |x'|^2 < \varrho^2\}$, then there exists some $r \in (0, \varrho)$ and a real analytic function $u = \sum_\alpha u_\alpha x^\alpha$ on B_r^n solving the above. Moroever, this solution is unique as an element of C^ω.

Proof. Non-examinable; see handout for proof. ∎

The proof is analogous to the ODE case, with the new step being the use of the PDE to find all derivatives. We illustrate this through the following example.

Example. Consider the first-order system

$$\begin{cases} u_t = v_x - f \\ v_t = -u_x \\ u(0) = 0 \\ v(0) = 0 \end{cases}$$

We compute the derivatives case by case:

Case 1: $\alpha = (k, 0)$

Differentiating the BC, we immediately get $\partial_x^k u(x, 0) = \partial_x^k v(x, 0) = 0$ for all k

Case 2: $\alpha = (k, 1)$

By the PDE, we have $u_t|_{t=0} = -f(x, 0)$ and $v_t|_{t=0} = 0$, so

$$\partial_x^k \partial_t u|_{t=0} = -\partial_x^k f(x, 0)$$
$$\partial_x^k \partial_t v|_{t=0} = 0$$

Case 3: $\alpha = (k, 2)$

By the PDE, we have $u_{tt} = v_{xt} - f_t$ and $v_{tt} = -u_{xt}$, so

$$\partial_x^k \partial_t^2 u|_{t=0} = -\partial_x^k \partial_t f|_{t=0}$$
$$\partial_x^k \partial_t^2 v|_{t=0} = \partial_x^{k+1} f|_{t=0}$$

The rest follow by induction.

It turns out that we can reduce a more general class of PDEs to the above first-order form. Consider the kth order scalar quasilinear PDE

$$\sum_{|\alpha|=k} a_\alpha(x, u, \cdots, D^{k-1}u) D^\alpha u + a_0(x, u, \cdots, D^{k-1}u) = 0$$

on some domain $\mathcal{U} \subset \mathbb{R}^n$.